

Advanced Calculus HW 9

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Question 1

(a)

$$\begin{aligned}\int_1^2 \int_0^y xy^2 dx dy &= \int_1^2 \frac{y^4}{2} dy \\ &= \frac{1}{10}(2^5 - 1) \\ &= 3.1\end{aligned}$$

$$\begin{aligned}\int_0^1 \int_1^2 xy^2 dy dx + \int_1^2 \int_x^2 xy^2 dy dx &= \int_0^1 x\left(\frac{7}{3}\right) dx + \int_1^2 x\left(\frac{8}{3} - \frac{x^3}{3}\right) dx \\ &= \frac{7}{6} + \frac{16}{5} - \frac{19}{15} \\ &= 3.1\end{aligned}$$

(b)

$$\begin{aligned}\int_0^5 \int_{5-x}^{\sqrt{25-x^2}} y dy dx &= \frac{1}{2} \int_0^5 (25 - x^2 - 25 + 10x - x^2) dx \\ &= \frac{125}{6}\end{aligned}$$

$$\begin{aligned}\int_0^5 \int_{5-y}^{\sqrt{25-y^2}} y dx dy &= \int_0^5 y(\sqrt{25-y^2} - 5 + y) dy \\ &= \left[-\frac{1}{3}(25-y^2)^{\frac{3}{2}} - \frac{5}{2}y^2 + \frac{y^3}{3} \right]_0^5 \\ &= \frac{125}{6}\end{aligned}$$

Question 2

Note, the plane $x + z = 1$ intersects the other plane at the line $y = 1$,
For the second plane, $y \geq 0$ and $y \leq 1$ in the region before $y = 1$

$$\begin{aligned}\int_0^1 \int_{-\sqrt{x}}^{\sqrt{x}} (1-x) dy dx &= \int_0^1 2(\sqrt{x} - x^{\frac{3}{2}}) dx \\ &= \frac{8}{15}\end{aligned}$$

Question 3

$$\begin{aligned}\int_0^1 \int_0^{2x} \cos(x^2) dy dx &= \int_0^1 2x \cos(x^2) dx \\ &= [\sin(x^2)]_0^1 \\ &= \sin(1)\end{aligned}$$

Question 4

$\sqrt{1-x^2-y^2}$ is a unit hemisphere

The region we are integrating over is a unit circle in the first quadrant

Hence the volume above this region is simply an eighth of the volume of a unit sphere

$$\frac{1}{8} \left(\frac{4}{3} \pi \right) = \frac{\pi}{6}$$